Vectors I Basics Worksheet: Solutions

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Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product.

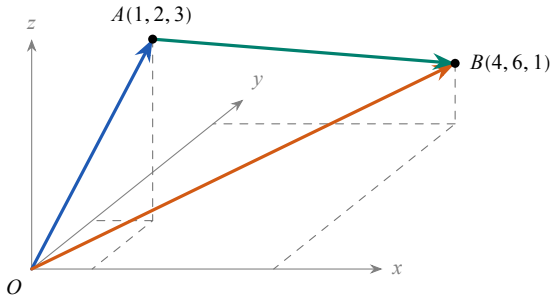
1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a}=3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$ as a column vector, and write $\mathbf{b}=\begin{pmatrix}4\\0\\-1\end{pmatrix}$ in terms of **i**, **j** and **k**.

$$\mathbf{a}=\begin{pmatrix}3\\-2\\5\end{pmatrix}$$

$$\begin{aligned}\mathbf{b}&=4\mathbf{i}+0\mathbf{j}-1\mathbf{k}\\&=4\mathbf{i}-\mathbf{k}\end{aligned}$$

2 The points A and B have coordinates $(1,2,3)$ and $(4,6,1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



$$\begin{aligned}\overrightarrow{AB}&=\overrightarrow{OB}-\overrightarrow{OA}\\&=\begin{pmatrix}4\\6\\1\end{pmatrix}-\begin{pmatrix}1\\2\\3\end{pmatrix}\\&=\begin{pmatrix}3\\4\\-2\end{pmatrix}\\&=3\mathbf{i}+4\mathbf{j}-2\mathbf{k}\end{aligned}$$

3 Given $\mathbf{p}=2\mathbf{i}+\mathbf{k}$ and $\mathbf{q}=-\mathbf{i}+3\mathbf{j}$, find **r** such that $2\mathbf{p}-3\mathbf{q}+\mathbf{r}=\mathbf{0}$. Give **r** in terms of **i**, **j** and **k**.

$$\mathbf{p}=\begin{pmatrix}2\\0\\1\end{pmatrix},\quad \mathbf{q}=\begin{pmatrix}-1\\3\\0\end{pmatrix}$$

$$\begin{aligned}\mathbf{r}&=3\mathbf{q}-2\mathbf{p}\\&=3\begin{pmatrix}-1\\3\\0\end{pmatrix}-2\begin{pmatrix}2\\0\\1\end{pmatrix}\\&=\begin{pmatrix}-3\\9\\0\end{pmatrix}-\begin{pmatrix}4\\0\\2\end{pmatrix}\\&=\begin{pmatrix}-7\\9\\-2\end{pmatrix}\\&=-7\mathbf{i}+9\mathbf{j}-2\mathbf{k}\end{aligned}$$

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$	**T**
(b)	The **j**-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1 $4\mathbf{i}-\mathbf{j}+2\mathbf{k}=\begin{pmatrix}4\\-1\\2\end{pmatrix}$	**T**
(c)	The position vector of $P(1,-2,0)$ is $\mathbf{i}-2\mathbf{j}$ $\overrightarrow{OP}=\begin{pmatrix}1\\-2\\0\end{pmatrix}=\mathbf{i}-2\mathbf{j}$	**T**
(d)	$3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$ $3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\0\\2\end{pmatrix}\neq\begin{pmatrix}3\\2\\0\end{pmatrix}$	**F**
(e)	$\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$ $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$	**F**

2 Magnitude

1 Find $|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}|$.

$$\begin{aligned}|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}| &= \sqrt{2^2+(-3)^2+6^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix}3\\-4\\2\end{pmatrix}-\begin{pmatrix}1\\2\\-1\end{pmatrix} \\ &= \begin{pmatrix}2\\-6\\3\end{pmatrix}\end{aligned}$$

$$\begin{aligned}AB = \left|\overrightarrow{AB}\right| &= \sqrt{2^2+(-6)^2+3^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

3 Given $\mathbf{a}=\begin{pmatrix}t\\2\\t-1\end{pmatrix}$ and $|\mathbf{a}|=3$, find the possible values of t.

$$\begin{aligned}|\mathbf{a}|^2 &= t^2+2^2+(t-1)^2 \\ 9 &= t^2+4+t^2-2t+1 \\ 0 &= 2t^2-2t-4 \\ 0 &= t^2-t-2 \\ 0 &= (t-2)(t+1) \\ t &= 2 \text{ or } t=-1\end{aligned}$$

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\lvert-\mathbf{a}\rvert=\lvert\mathbf{a}\rvert$ for every vector $\mathbf{a}$ $\lvert-\mathbf{a}\rvert=\lvert-1\rvert\,\lvert\mathbf{a}\rvert=\lvert\mathbf{a}\rvert$	**T**
(b)	$\lvert3\mathbf{a}\rvert=3\lvert\mathbf{a}\rvert$ $\lvert3\mathbf{a}\rvert=\lvert3\rvert\,\lvert\mathbf{a}\rvert=3\lvert\mathbf{a}\rvert$	**T**
(c)	$\left\lvert\begin{pmatrix}0\\-5\\0\end{pmatrix}\right\rvert=-5$ $\left\lvert\begin{pmatrix}0\\-5\\0\end{pmatrix}\right\rvert=\sqrt{0^2+(-5)^2+0^2}=5$	**F**

	Statement	**T/F**
(d)	$\left\|\begin{pmatrix}1\\1\\1\end{pmatrix}\right\| = 1$ $\left\|\begin{pmatrix}1\\1\\1\end{pmatrix}\right\| = \sqrt{1^2+1^2+1^2} = \sqrt{3} \neq 1$	**F**
(e)	$\|\mathbf{i}+2\mathbf{j}-2\mathbf{k}\| = 1+2-2 = 1$ $\|\mathbf{i}+2\mathbf{j}-2\mathbf{k}\| = \sqrt{1^2+2^2+(-2)^2} = 3$	**F**
(f)	$\|\mathbf{a}+\mathbf{b}\| = \|\mathbf{a}\|+\|\mathbf{b}\|$ for all vectors $\mathbf{a}$, $\mathbf{b}$ $\|\mathbf{i}+\mathbf{j}\| = \sqrt{2} \neq \|\mathbf{i}\|+\|\mathbf{j}\| = 2$	**F**
(g)	$\|\mathbf{a}+\mathbf{b}\| \leq \|\mathbf{a}\|+\|\mathbf{b}\|$ for all vectors $\mathbf{a}$, $\mathbf{b}$ $\overrightarrow{OA} = \mathbf{a},\ \overrightarrow{AB} = \mathbf{b},\ \overrightarrow{OB} = \mathbf{a}+\mathbf{b},\ OB \leq OA+AB\ (\triangle OAB)$	**T**
(h)	For all $\lambda \in \mathbb{R}$, $\|\lambda\mathbf{a}\| = \lambda\|\mathbf{a}\|$ $\lambda = -1,\ \mathbf{a} = \mathbf{i}:\ \|-\mathbf{i}\| = 1 \neq -1;\ \|\lambda\mathbf{a}\| = \|\lambda\|\|\mathbf{a}\|$	**F**

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

$$\left|\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}\right| = \sqrt{2^2+(-1)^2+2^2}$$
$$= 3$$
$$\text{unit vector} = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

$$\left|\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}\right| = \sqrt{3^2+0^2+(-4)^2}$$
$$= 5$$
$$\text{vector} = 10 \cdot \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$$
$$= \begin{pmatrix} 6 \\ 0 \\ -8 \end{pmatrix}$$

3 The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$\left|\overrightarrow{AB}\right| = \sqrt{2^2+(-6)^2+3^2}$$
$$= 7$$
$$\text{vector} = -14 \cdot \frac{1}{7}\begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} -4 \\ 12 \\ -6 \end{pmatrix}$$

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector $\left\lvert\begin{pmatrix}1\\1\\1\end{pmatrix}\right\rvert = \sqrt{3} \neq 1$	**F**
(b)	$\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector $\left\lvert\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}\right\rvert = \frac{1}{3}\sqrt{1+4+4} = 1$	**T**
(c)	$\mathbf{i}+\mathbf{j}$ is a unit vector $\lvert\mathbf{i}+\mathbf{j}\rvert = \sqrt{1^2+1^2+0^2} = \sqrt{2} \neq 1$	**F**
(d)	$\lvert\hat{\mathbf{a}}\rvert = 1$ for every non-zero vector $\mathbf{a}$ $\lvert\hat{\mathbf{a}}\rvert = \left\lvert\frac{\mathbf{a}}{\lvert\mathbf{a}\rvert}\right\rvert = \frac{\lvert\mathbf{a}\rvert}{\lvert\mathbf{a}\rvert} = 1$	**T**
(e)	$\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel $\mathbf{a} = \lvert\mathbf{a}\rvert\,\hat{\mathbf{a}}$	**T**

4 Parallel Vectors ($\mathbf{a} = \lambda \mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

$$\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$$
$$= -3 \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$
$$= -3\mathbf{a}$$
$$\therefore \mathbf{a} \parallel \mathbf{b} \text{ (shown)}$$

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

$$\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix} = \lambda \begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$$
$$4 = -2\lambda$$
$$\lambda = -2$$
$$p = 3\lambda$$
$$= -6$$
$$-2 = \lambda q$$
$$-2 = -2q$$
$$q = 1$$

3 Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

$$|\mathbf{a}| = \sqrt{1^2 + 2^2 + (-2)^2}$$
$$= 3$$
$$\text{vector} = \pm 6 \cdot \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$
$$= \pm \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$$

	Statement	**T/F**

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$ $2\mathbf{i} \parallel \mathbf{i},\ 2\mathbf{i} \neq \mathbf{i}$	**F**
(b)	$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$ $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$	**T**
(c)	$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ $\frac{2}{1} = \frac{4}{2} \neq \frac{6}{4}$	**F**
(d)	$\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$ $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix} = -3 \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$	**T**
(e)	If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions $\mathbf{a} = \lambda \mathbf{b},\ \lambda = -2$	**F**

5 Collinear Points

1 Show that the points $A(1,2,3)$, $B(3,5,4)$ and $C(7,11,6)$ are collinear.

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$
$$\overrightarrow{BC} = \begin{pmatrix} 7 \\ 11 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix}$$
$$= \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix}$$
$$= 2\overrightarrow{AB}$$
$$\therefore \overrightarrow{AB} \parallel \overrightarrow{BC},\ B \text{ common}$$
$$\therefore A, B, C \text{ collinear (shown)}$$

2 For the points in Question 1, find the ratio $AB : BC$.

$$\overrightarrow{BC} = 2\overrightarrow{AB}$$
$$BC = 2AB$$
$$AB : BC = 1 : 2$$

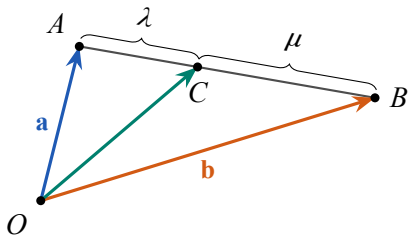
3 The points $P(2,-1,0)$, $Q(4,1,3)$ and $R(a,7,b)$ are collinear. Find a and b.

$$\overrightarrow{PR} = \lambda\overrightarrow{PQ}$$
$$\begin{pmatrix} a-2 \\ 8 \\ b \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$
$$8 = 2\lambda$$
$$\lambda = 4$$
$$a - 2 = 2(4)$$
$$a = 10$$
$$b = 3(4)$$
$$= 12$$

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear $A(0,0,0), B(1,0,0), C(0,1,0), D(1,1,0) : \overrightarrow{AB} = \overrightarrow{CD}$	**F**
(b)	If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear $\overrightarrow{AB} \parallel \overrightarrow{AC},\ A$ common	**T**
(c)	$A(0,0,0)$, $B(1,1,1)$ and $C(2,2,3)$ are collinear $\overrightarrow{AC} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \neq \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$	**F**
(d)	The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$) $2\mathbf{a} - \mathbf{a} = \mathbf{a},\ -\mathbf{a} - \mathbf{a} = -2(\mathbf{a})$	**T**

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC:CB=\lambda:\mu$, $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$.

1 Let $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

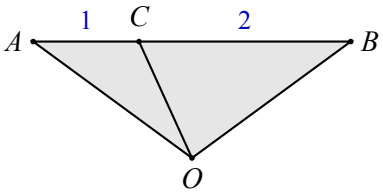
(a) $AC:CB=1:2$,

(b) $3AC=2CB$,

(c) $2AB=5CB$,

(d) Given instead that C lies on AB produced, and $3AC=7BC$.

(a)

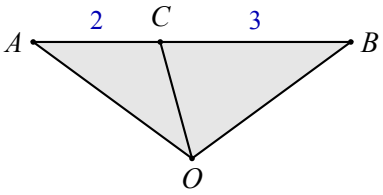


$$\overrightarrow{OC}=\frac{2\overrightarrow{OA}+\overrightarrow{OB}}{3}$$

$$=\begin{pmatrix}1\\ \frac{4}{3}\\ \frac{5}{3}\end{pmatrix}$$

$$\therefore C\left(1,\tfrac{4}{3},\tfrac{5}{3}\right)$$

(b)



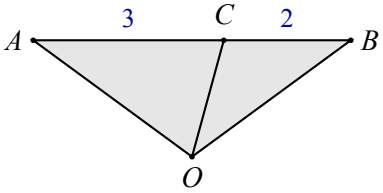
$$\frac{AC}{CB}=\frac{2}{3}$$

$$\overrightarrow{OC}=\frac{2\overrightarrow{OB}+3\overrightarrow{OA}}{5}$$

$$=\begin{pmatrix}1\\ \frac{7}{5}\\ \frac{9}{5}\end{pmatrix}$$

$$\therefore C\left(1,\tfrac{7}{5},\tfrac{9}{5}\right)$$

(c)



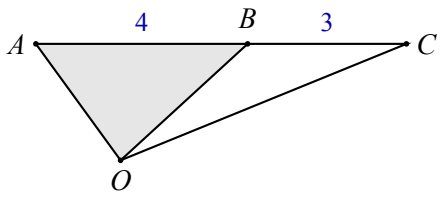
$$\frac{AB}{CB}=\frac{5}{2}$$

$$\overrightarrow{OC}=\frac{3\overrightarrow{OB}+2\overrightarrow{OA}}{5}$$

$$=\begin{pmatrix}1\\ \frac{8}{5}\\ \frac{11}{5}\end{pmatrix}$$

$$\therefore C\left(1,\tfrac{8}{5},\tfrac{11}{5}\right)$$

(d)



$$\frac{AC}{BC}=\frac{7}{3}$$
$$\overrightarrow{OB}=\frac{3\overrightarrow{OA}+4\overrightarrow{OC}}{7}$$
$$\overrightarrow{OC}=\frac{7\overrightarrow{OB}-3\overrightarrow{OA}}{4}$$
$$=\begin{pmatrix}1\\ \frac{11}{4}\\ \frac{9}{2}\end{pmatrix}$$
$$\therefore C\left(1,\tfrac{11}{4},\tfrac{9}{2}\right)$$

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3,-1,5)$ and $(-1,5,1)$.

$$\overrightarrow{OM}=\frac{1}{2}\left[\begin{pmatrix}3\\-1\\5\end{pmatrix}+\begin{pmatrix}-1\\5\\1\end{pmatrix}\right]$$
$$=\frac{1}{2}\begin{pmatrix}2\\4\\6\end{pmatrix}$$
$$=\begin{pmatrix}1\\2\\3\end{pmatrix}$$

3 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\overrightarrow{OC}=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$ $\overrightarrow{OC}=\mathbf{a}+\frac{\lambda}{\lambda+\mu}(\mathbf{b}-\mathbf{a})=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$	**T**
(b)	$\overrightarrow{OC}=\frac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$ $\overrightarrow{OC}=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$	**F**
(c)	If $\lambda:\mu=1:1$, then $\overrightarrow{OC}=\frac{1}{2}(\mathbf{a}+\mathbf{b})$ $\overrightarrow{OC}=\frac{1\mathbf{a}+1\mathbf{b}}{2}$	**T**
(d)	If $\lambda:\mu=1:3$, then $\overrightarrow{OC}=\frac{3\mathbf{a}+\mathbf{b}}{4}$ $\overrightarrow{OC}=\frac{3\mathbf{a}+1\mathbf{b}}{1+3}$	**T**
(e)	If $\lambda:\mu=2:1$, then $\overrightarrow{OC}=\frac{2\mathbf{a}+\mathbf{b}}{3}$ $\overrightarrow{OC}=\frac{1\mathbf{a}+2\mathbf{b}}{3}$	**F**

7 Scalar Product

1 Properties of scalar product

It is given that $\mathbf{a}=\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

(a) $\mathbf{a}\cdot\mathbf{b}$ (b) $\mathbf{b}\cdot\mathbf{a}$ (c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$ (d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$

(e) $\mathbf{a}\cdot(2\mathbf{c})$ (f) $(2\mathbf{a})\cdot\mathbf{c}$ (g) $\mathbf{a}\cdot\mathbf{a}$

Ans: (a) 22 (b) 22 (c) 13 (d) 13 (e) -18 (f) -18 (g) 14

2 Scalar product formula: $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$

(a) Points O, A, B are such that $\overrightarrow{OA}=\mathbf{i}+4\mathbf{j}-3\mathbf{k}$ and $\overrightarrow{OB}=5\mathbf{i}-\mathbf{j}$

(i) Find the size of angle AOB

(ii) Find the size of angle OAB.

(b) It is given that $\mathbf{c}=\begin{pmatrix}2\\3\\6\end{pmatrix}$, $\mathbf{d}=\begin{pmatrix}1\\-2\\3\end{pmatrix}$ and $\mathbf{e}=\begin{pmatrix}3\\2\\-2\end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and

(i) the vector $\mathbf{d}$

(ii) the vector $\mathbf{e}$

(iii) the positive y-axis

(c) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:

$$|\mathbf{a}||\mathbf{b}|=6.$$

Find the value of $\mathbf{a}\cdot\mathbf{b}$ for each of the following scenarios.

(i) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $60°$.

(ii) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $120°$.

(iii) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Ans:

(a)(i) $87.8°$ (ii) $133.9°$

(b)(i) $57.7°$ (ii) $90°$ (iii) $64.6°$

(c)(i) $\mathbf{a}\cdot\mathbf{b}=3$ (ii) $\mathbf{a}\cdot\mathbf{b}=-3$ (iii) $\mathbf{a}\cdot\mathbf{b}=0$

3 Perpendicularity of vectors

(a) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $2\mathbf{i}+4\mathbf{j}+\mathbf{k}$ respectively. Show that OA is perpendicular to OB.

(b) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a}=2p\mathbf{i}+3p\mathbf{j}+6p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$$

where $p>0$. It is given that $|\mathbf{a}|=|\mathbf{b}|$.

(i) Find the exact value of p.

(ii) Show that the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$ are perpendicular.

Ans: (b)(i) $p=\frac{3}{7}$

4 Length of projection

(a) Given that $\mathbf{a}=2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+2\mathbf{k}-\mathbf{k}$, find the length of projection and the projection vector of
   (i) **a** onto **b**,
   (ii) **b** onto **a**.

(b) With reference to the origin O, the points A, B and C have position vectors given by

$$\mathbf{a}=\mathbf{i}+2\mathbf{j}-\mathbf{k},\ \mathbf{b}=3\mathbf{i}+\mathbf{k} \text{ and } \mathbf{c}=2\mathbf{i}+2\mathbf{j}+3\mathbf{k}$$

respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

Ans:

(a)(i) $\frac{2}{\sqrt{6}}, \frac{1}{3}\begin{pmatrix}-1\\2\\-1\end{pmatrix}$ (ii) $\frac{2}{3}, \frac{2}{9}\begin{pmatrix}2\\1\\2\end{pmatrix}$

(b) $\frac{2}{3}$

8 Vector Product

1 Properties of vector product

(a) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

(b) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(i) $\mathbf{a} \times \mathbf{b}$; (ii) $\mathbf{b} \times \mathbf{a}$; (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (iv) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$;

(v) $\mathbf{a} \times (2\mathbf{b})$; (vi) $(2\mathbf{a}) \times \mathbf{b}$; (vii) $\mathbf{a} \times \mathbf{a}$;

Ans:

(a) $\begin{pmatrix} -29 \\ 17 \\ 26 \end{pmatrix}$

(b)(i) $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ (ii) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ (iii) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ (iv) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ (v) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ (vi) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ (vii) $\mathbf{0}$

2 Parallel vectors

(a) Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.

(b) Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.

(i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$

(ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$

(iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

Ans: (b) Neither; Perpendicular; Parallel

3 Vector product formula

Using the Vector Product, find the acute angles between the following vectors:

(a) $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ (b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ (c) $\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

Ans: (a) 51.0° (b) 46.4° (c) 87.0°

4 Perpendicular distance

(a) Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. Find the following:

(i) The shortest distance from A to OB

(ii) The shortest distance from B to OA

(b) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ respectively.

(i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.

(ii) Hence, or otherwise, find the shortest distance from B to OA.

(c) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$, $\mathbf{b}=\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $\mathbf{c}=3\mathbf{i}-\mathbf{j}+\mathbf{k}$. Find the shortest distance from C to the line joining A and B.

Ans:

(a)(i) 4.09 units (ii) 4.51 units

(b)(i) $2\sqrt{3}$ units (ii) $\sqrt{2}$ units

(c) 2 units

5 Areas of triangles & parallelograms

(a) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA}=\begin{pmatrix}1\\0\\3\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}-2\\3\\2\end{pmatrix}$. Find the area of triangle OAB.

(b) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA}=\begin{pmatrix}0\\1\\1\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}1\\-2\\2\end{pmatrix}$. Find the area of triangle OAB.

(c) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are

$$\mathbf{i}+\mathbf{j}-\mathbf{k},\ 2\mathbf{i}+\mathbf{k} \text{ and } -\mathbf{i}+2\mathbf{j}-\mathbf{k}$$

respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find the exact area of triangle ABC.

Ans: (a) $\frac{\sqrt{154}}{2}$ units^2 (b) $\frac{3\sqrt{2}}{2}$ units^2 (c) $\frac{\sqrt{21}}{2}$ units^2

9 Practice Problems

1 Relative to the origin O, the position vectors of two points A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively. The length of $\mathbf{a}$ is 4 units and $\mathbf{b}$ is a unit vector. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{3}$ radians.

(a) Find the area of triangle OAB. [2]

It is given that the point M lies on the line segment AB such that $AM:AB$ is $3:4$.

(b) Find the exact area of triangle OAM. [3]

It is given that the point C lies on AB produced, such that $\overrightarrow{AC}=\lambda\overrightarrow{AB}$, where $\lambda>1$.

(c) Find the value of λ such that OC is perpendicular to AB. [4]

Solution

(a)

$$\begin{aligned}\text{Area of } \Delta OAB &= \tfrac{1}{2}|\mathbf{a}\times\mathbf{b}| \\ &= \tfrac{1}{2}|\mathbf{a}||\mathbf{b}|\sin\tfrac{\pi}{3} \\ &= \tfrac{1}{2}(4)(1)\left(\tfrac{\sqrt{3}}{2}\right) \\ &= \sqrt{3}\text{ units}^2\end{aligned}$$

(b)

$$\begin{aligned}AM:MB &= 3:1 \\ \overrightarrow{OM} &= \tfrac{1\cdot\mathbf{a}+3\cdot\mathbf{b}}{4} = \tfrac{1}{4}\mathbf{a}+\tfrac{3}{4}\mathbf{b} \\ \text{Area of } \Delta OAM &= \tfrac{1}{2}\left|\mathbf{a}\times\left(\tfrac{1}{4}\mathbf{a}+\tfrac{3}{4}\mathbf{b}\right)\right| \\ &= \tfrac{1}{2}\left|\tfrac{1}{4}(\mathbf{a}\times\mathbf{a})+\tfrac{3}{4}(\mathbf{a}\times\mathbf{b})\right| \\ &= \tfrac{3}{4}\cdot\tfrac{1}{2}|\mathbf{a}\times\mathbf{b}| \quad (\mathbf{a}\times\mathbf{a}=\mathbf{0}) \\ &= \tfrac{3}{4}\sqrt{3} \\ &= \tfrac{3\sqrt{3}}{4}\text{ units}^2\end{aligned}$$

(c)

$$\begin{aligned}\mathbf{a}\cdot\mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\tfrac{\pi}{3} = (4)(1)\tfrac{1}{2} = 2 \\ \overrightarrow{OC} &= \mathbf{a}+\lambda(\mathbf{b}-\mathbf{a}) \\ \overrightarrow{OC}\cdot\overrightarrow{AB} &= 0 \\ [\mathbf{a}+\lambda(\mathbf{b}-\mathbf{a})]\cdot(\mathbf{b}-\mathbf{a}) &= 0 \\ \mathbf{a}\cdot\mathbf{b}-|\mathbf{a}|^2+\lambda(|\mathbf{b}|^2-2\mathbf{a}\cdot\mathbf{b}+|\mathbf{a}|^2) &= 0 \\ 2-16+\lambda(1-4+16) &= 0 \\ -14+13\lambda &= 0 \\ \lambda &= \frac{14}{13}\end{aligned}$$

2 The non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy $|\mathbf{a}|=3$, $|\mathbf{b}|=4$ and $\mathbf{a}\cdot\mathbf{b}=6$.

(a) Find the angle between $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Find $|\mathbf{a}-\mathbf{b}|$. [3]

Solution

(a)

$$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{6}{3\cdot4}=\frac{1}{2}$$
$$\theta=\frac{\pi}{3}\quad(\text{or } 60^\circ)$$

(b)

$$\begin{aligned}|\mathbf{a}-\mathbf{b}|^2&=(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})\\&=|\mathbf{a}|^2-2\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2\\&=9-2(6)+16\\&=13\\|\mathbf{a}-\mathbf{b}|&=\sqrt{13}\end{aligned}$$

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3 Relative to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $3\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel.

(a) Find the area of triangle OAC in terms of $|\mathbf{a}\times\mathbf{b}|$. [2]

(b) The point D is the midpoint of AC. Find the area of triangle OAD in terms of $|\mathbf{a}\times\mathbf{b}|$. [2]

Solution

(a)

$$\begin{aligned}\text{Area of }\Delta OAC&=\tfrac{1}{2}|\mathbf{a}\times(3\mathbf{b})|\\&=\tfrac{1}{2}\cdot3\,|\mathbf{a}\times\mathbf{b}|\\&=\frac{3}{2}|\mathbf{a}\times\mathbf{b}|\end{aligned}$$

(b)

$$\overrightarrow{OD}=\tfrac{1}{2}(\mathbf{a}+3\mathbf{b})$$
$$\begin{aligned}\text{Area of }\Delta OAD&=\tfrac{1}{2}\left|\mathbf{a}\times\tfrac{1}{2}(\mathbf{a}+3\mathbf{b})\right|\\&=\tfrac{1}{4}\,|\mathbf{a}\times\mathbf{a}+3(\mathbf{a}\times\mathbf{b})|\\&=\tfrac{1}{4}\cdot3\,|\mathbf{a}\times\mathbf{b}|\\&=\frac{3}{4}|\mathbf{a}\times\mathbf{b}|\end{aligned}$$

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4 The non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $3\mathbf{a}\times\mathbf{b}=2\mathbf{c}\times\mathbf{b}$.

(a) Show that $3\mathbf{a}-2\mathbf{c}=\lambda\mathbf{b}$ for some scalar λ. [2]

(b) It is now given that $|\mathbf{a}|=2$, $|\mathbf{b}|=3$ and the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{6}$. Find $|3\mathbf{a}-2\mathbf{b}|$. [3]

Solution

(a)

$$3\mathbf{a}\times\mathbf{b}-2\mathbf{c}\times\mathbf{b}=\mathbf{0}$$
$$(3\mathbf{a}-2\mathbf{c})\times\mathbf{b}=\mathbf{0}$$

Since $\mathbf{b}\neq\mathbf{0}$, the vector $3\mathbf{a}-2\mathbf{c}$ is parallel to $\mathbf{b}$. Hence $3\mathbf{a}-2\mathbf{c}=\lambda\mathbf{b}$ for some scalar λ.

(b)

$$\begin{aligned}\mathbf{a}\cdot\mathbf{b}&=|\mathbf{a}||\mathbf{b}|\cos\tfrac{\pi}{6}=(2)(3)\tfrac{\sqrt{3}}{2}=3\sqrt{3}\\ |3\mathbf{a}-2\mathbf{b}|^2&=(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})\\ &=9|\mathbf{a}|^2-12\mathbf{a}\cdot\mathbf{b}+4|\mathbf{b}|^2\\ &=9(4)-12(3\sqrt{3})+4(9)\\ &=36-36\sqrt{3}+36\\ &=72-36\sqrt{3}\\ |3\mathbf{a}-2\mathbf{b}|&=\sqrt{36(2-\sqrt{3})}\\ &=6\sqrt{2-\sqrt{3}}\end{aligned}$$

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5 Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}|=2$ and that $\mathbf{b}$ is a unit vector. The point C divides the line segment AB internally such that $AC:CB=3:2$. The point D lies on OB produced such that $\overrightarrow{OD}=3\overrightarrow{OB}$.

(a) Find $\overrightarrow{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Find $\overrightarrow{OD}$ in terms of $\mathbf{b}$. [1]

(c) Given that CD is perpendicular to AB, find the value of $\mathbf{a}\cdot\mathbf{b}$. [3]

Solution

(a)

$$AC:CB=3:2$$
$$\overrightarrow{OC}=\frac{2\mathbf{a}+3\mathbf{b}}{5}=\frac{2}{5}\mathbf{a}+\frac{3}{5}\mathbf{b}$$

(b)

$$\overrightarrow{OD}=3\mathbf{b}$$

(c)

$$\begin{aligned}
\overrightarrow{CD} &= \overrightarrow{OD} - \overrightarrow{OC} = 3\mathbf{b} - \left(\tfrac{2}{5}\mathbf{a} + \tfrac{3}{5}\mathbf{b}\right) = -\tfrac{2}{5}\mathbf{a} + \tfrac{12}{5}\mathbf{b} \\
\overrightarrow{AB} &= \mathbf{b} - \mathbf{a} \\
\overrightarrow{CD} \cdot \overrightarrow{AB} &= 0 \\
\left(-\tfrac{2}{5}\mathbf{a} + \tfrac{12}{5}\mathbf{b}\right) \cdot (\mathbf{b} - \mathbf{a}) &= 0 \\
-\tfrac{2}{5}\mathbf{a}\cdot\mathbf{b} + \tfrac{2}{5}|\mathbf{a}|^2 + \tfrac{12}{5}|\mathbf{b}|^2 - \tfrac{12}{5}\mathbf{a}\cdot\mathbf{b} &= 0 \\
\tfrac{2}{5}(4) + \tfrac{12}{5}(1) - \left(\tfrac{2}{5} + \tfrac{12}{5}\right)\mathbf{a}\cdot\mathbf{b} &= 0 \\
\tfrac{8}{5} + \tfrac{12}{5} - \tfrac{14}{5}\mathbf{a}\cdot\mathbf{b} &= 0 \\
4 &= \tfrac{14}{5}\mathbf{a}\cdot\mathbf{b} \\
\mathbf{a}\cdot\mathbf{b} &= \frac{10}{7}
\end{aligned}$$

END OF WORKSHEET